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Magnetostratigraphy of Early–Middle Toarcian expanded sections from the Iberian Range (central Spain)

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Abstract

The magnetostratigraphy of the Lower–Middle Toarcian has been established in two well-dated stratigraphically expanded sections: the Sierra Palomera and the Ariño sections, located in the Iberian Range, in central-eastern Spain. Two magnetisation components could be isolated by thermal cleaning: a secondary syntectonic component of always normal polarity unblocking at intermediate temperatures up to 450 °C/475 °C (A component) and a high temperature unblocking component up to 575 °C (B component). The B component passes fold and reversal tests and is considered the characteristic remanent magnetisation of primary origin. The first Toarcian palaeomagnetic pole for Iberia has been obtained: Plat=77.4°, Plon=241.3°E (dm=5.4° dp=6.0°).

Five pairs of normal and reversed polarity zones were calibrated to regional ammonite subzones. The pattern can be calibrated to other Toarcian magnetostratigraphic studies, but provides a more detailed biostratigraphic framework. A refined magnetic polarity time scale is proposed for the Lower–Middle Toarcian.
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Keywords: magnetostratigraphy; Early Jurassic; biostratigraphy; Iberian Plate

1. Introduction

Reversals of the Earth's magnetic field are recorded in the oceanic magnetic anomaly sequence since the Middle Jurassic (Ogg, 1995a,b; Gradstein et al., 2004). For older periods, the Magnetic Polarity Time Scale (MPTS) should be established from magnetostratigraphic studies of well-dated sedimentary sections on land.

Several magnetostratigraphic studies have been carried out during the last decades on Jurassic sedimentary sections in the Spanish Iberian and Betic Ranges (Ogg et al., 1984; Steiner et al., 1986, 1987; Galbrun et al., 1990; Juárez et al., 1994, 1995). Due to their suitable magnetic properties and good ammonite-based biostratigraphical control, some of

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these Iberian magnetostratigraphic columns have been used as reference sections for the construction of the Middle–Late Jurassic MPTS (Ogg, 1995a,b; Gradstein et al., 1994, 2004; Opdyke and Channell, 1996).

The MPTS of the Toarcian interval (183.0 to 175.5 Ma) (Gradstein et al., 2004; Ogg, 2004), is based on a small number of magnetostratigraphic studies, many of them carried out in condensed sections, for instance in the Southalpine Breggia section (Horner and Heller, 1983) and in the old stratotype of the Toarcian at Thouars (Paris Basin) (Galbrun et al., 1988a). The magnetostratigraphic section of Iznalloz (southern Spain) (Galbrun et al., 1990) was partially correlated with the existing MPTS. The Middle Toarcian magnetostratigraphy obtained from the Valdorbia section (Italy) (Channell et al., 1984) was not used in the previous MPTS. Recent results from the Neuquén basin (Argentina) (Iglesia-Llanos and Riccardi, 2000) give contradictory results. Hence it is suggested that a revision of the MPTS should be considered.

This paper presents a magnetostratigraphic study performed in two marine limestone sections located in the Iberian Range (Spain): the Sierra Palomera section (lat.=40.6°N, long.=−1.25°E) and the Ariño section (lat.=41.0°N, long.=−0.9°E). The sections were selected because 1) no indications of major sedimentary discontinuities were found, suggesting almost continuous sedimentation in the considered interval, 2) abundance and diversity of ammonites provides a high-resolution biostratigraphy, 3) the studied sections are characterized by high sedimentation rates, which are also taken as indicators of a rather complete sedimentary record, 4) they provide the possibility of a regional fold test, and 5) the palaeogeographic location of the Iberian Range allows for a better correlation between the European and Tethyan realms (Fig. 1a). Average strike and dip of strata at Sierra Palomera is N330°E 40°NE and at Ariño N135°E 15°SW.

2. Geological setting

The Sierra Palomera and Ariño sections are located in the central sector of the Iberian Range which is a NW–SE trending folded and thrusted belt, located in eastern Iberia (Fig. 1b, c). Deposition of Lower Jurassic sediments took place in an intraplate extensional basin during an interval of relative tectonic quiescence between the Permian–Triassic and the Late Jurassic–Lower Cretaceous rifting

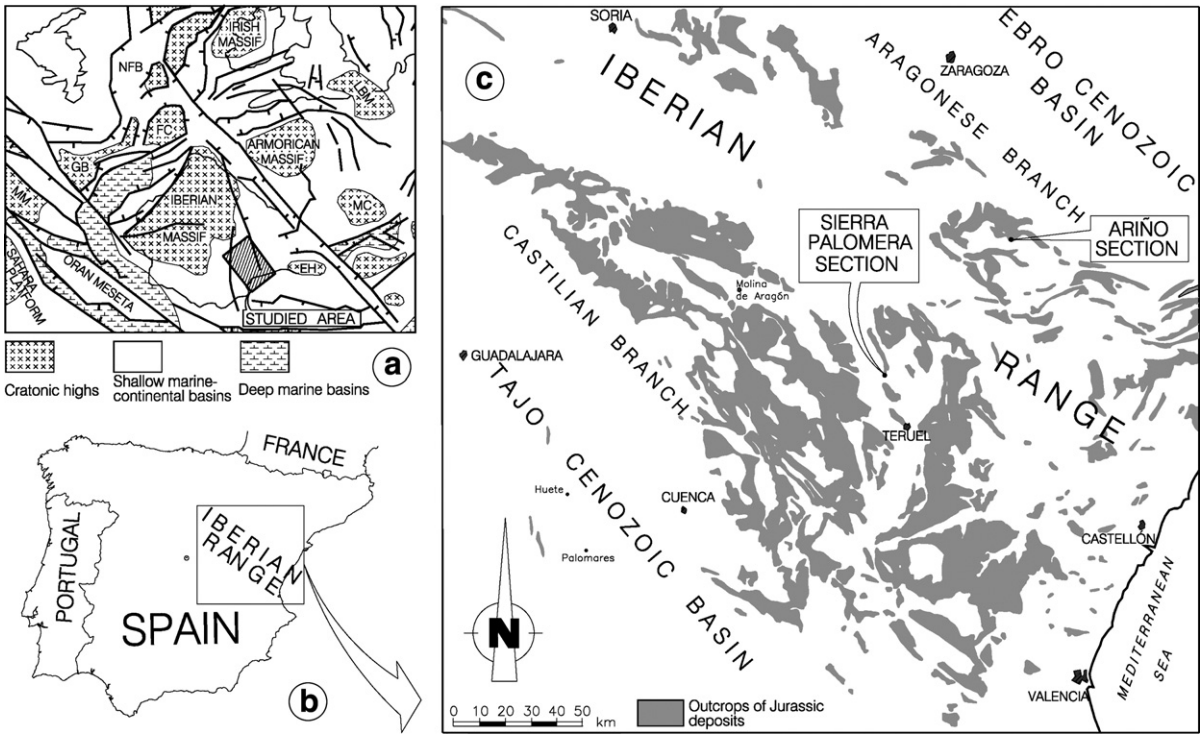


Fig. 1. Location maps of the studied sections. (a) Palaeogeographical map of Western Europe for the upper part of the Early Jurassic (modified after Ziegler, 1990), showing the situation of the studied area. NFB: East Newfoundland Basin. LBM: London-Brabant Massif. FC: Flemish Cap. GB: Grand Bank. MC: Massif Central High. MM: Morocco Meseta. EH: Ebro High. (b) Location of the Iberian Range within the Iberian Peninsula. (c) Outcrops of Jurassic deposits in the central and southern part of the Iberian Range and location of the Ariño and Sierra Palomera sections studied in this work.

episodes. During Toarcian, the Iberian platform system was characterized by an open marine external platform environment, where deposition of an alternation of limestones, predominantly grey mudstones, and marls, known as the Turmiel Formation took place (Goy et al., 1976; Gómez and Goy, 2000, 2004). Distribution of depocentres and morphological highs was controlled by rejuvenation of SW and NE trending Late-Variscan faults with associated local volcanic activity (Gómez and Goy, 2004, 2005) (Fig. 2). Depositional rates in the sections studied in the Iberian Range are much higher than in other sections studied in NW Europe or the Alps. Total thickness of the Toarcian deposits in the sections studied here is about 70 m. Assuming a total duration for the Toarcian of 9.5 My (189.6 to 180.1 Ma) (Gradstein et al., 1995) or 7.4 My (183.0 to 175.6 Ma) (Gradstein et al., 2004), and a constant sedimentation rate, average values are in the order of 7.4 or 9.5 m/My, respectively. However, depositional rate was not uniform over time. Sedimentation rate is much higher during the Early Toarcian in Ariño (7.8–10 m/My) than in Sierra Palomera (4.9–6.3 m/My). This tendency became inverted during the Middle Toarcian.

3. Biostratigraphy

Biostratigraphic analysis of the Sierra Palomera and the Ariño sections has been based on a bed by bed study and sampling of both sections and on the detailed determination of the collected ammonites. A complete and precise local biozonation has been previously published (Goy et al., 1988, 1997; Comas-Rengifo et al., 1996; García Joral and Goy, 2004; Gómez and Goy, 2004, 2005). In Figs. 3 and 4, a simplified ammonite-based biostratigraphy showing the distribution of the species characterizing the different zones and subzones is presented. The obtained biozonation has been compared and correlated with the standard scales

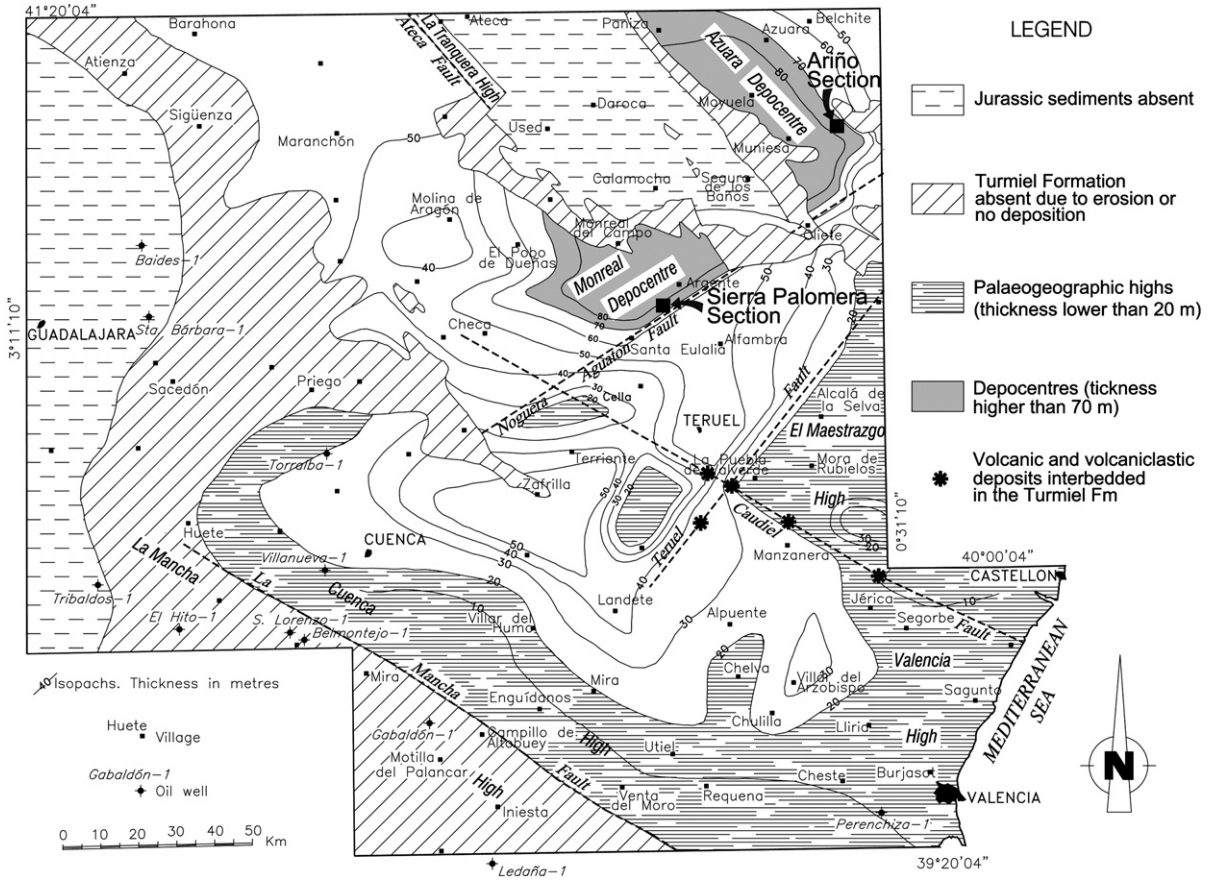


Fig. 2. Palaeogeographic elements active during deposition of the alternating limestones and marls of the Turmiel Formation (Toarcian in age, but includes locally part of the Late Pliensbachian and the Aalenian) in the central and southern portion of the Iberian platform system (modified after Gómez and Goy, 2005). The Sierra Palomera and Ariño sections are included in the main depocentres of the Toarcian deposits (Monreal and Azuara depocentres, respectively). External platform environments and sediment thickness are partly controlled by normal faulting, which was locally associated with volcaniclastic deposits.

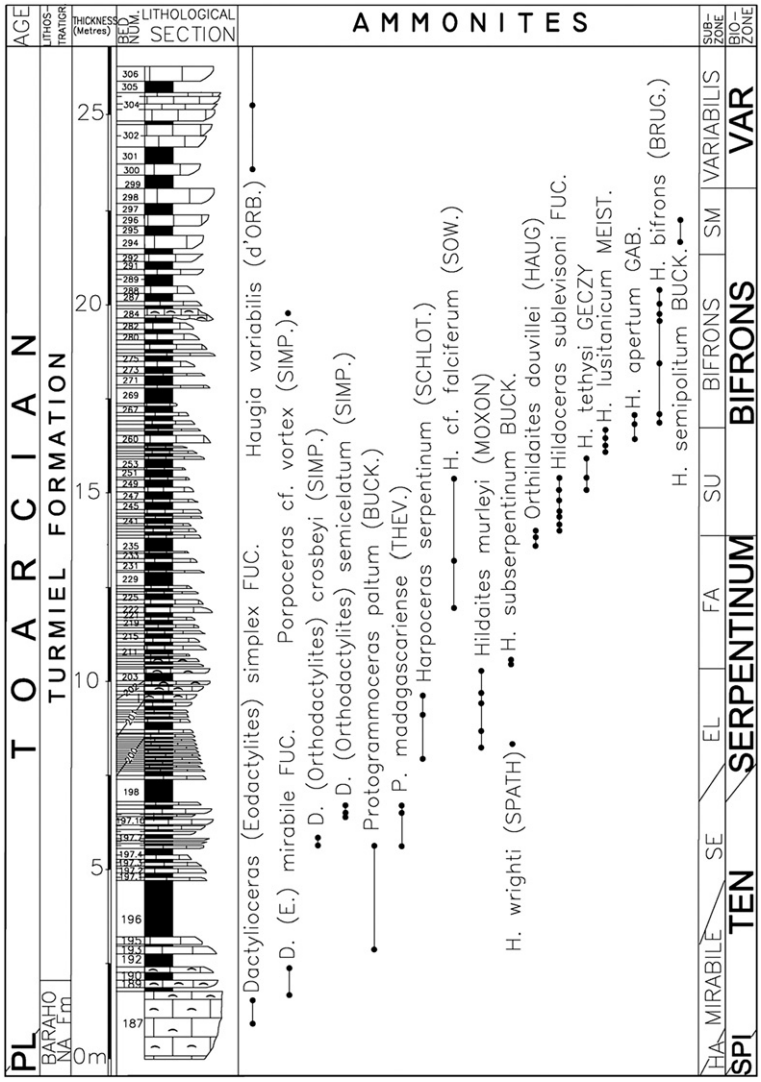


Fig. 3. Stratigraphic succession of the Sierra Palomera section showing the distribution of ammonites index characterizing the zones and subzones. These zones and subzones have been referred to the standard chronozones accepted in Europe. TEN: Tenuicostatum. VAR: Variabilis. HA: Hawskerense. SE: Semicelatum. EL: Elegantulum. FA: Falciferum. SU: Sublevisoni. SM: Semipolitum. Inclined line indicates the uncertainty interval of the zones and subzone boundaries.

(Dean et al., 1961; Elmi et al., 1994, 1997; Gradstein et al., 1995; Macchioni and Cecca, 2002; Page, 2003) and with local scales used in previous magnetostratigraphic studies (Gallitelli Wendt, 1969; Wiedenmayer, 1980; Braga et al., 1982; Goy et al., 1988; Goy and Martínez, 1990). It is shown in Fig. 5.

The lower boundary of the Toarcian has been marked with the First Appearance Datum (FAD) of the genus *Dactylioceras* (levels 187 at Sierra Palomera and 40 at Ariño), which in Ariño coincides with the FAD of *Protogrammoceras paltum*. During the Tenuicostatum Zone the species of *D. (Eodactylites): simplex-mirabile*, and *D. (Orthodactylites): crosbeyi–tenuicostatum–semicelatum* are in consecutive succession. This sequence and the presence of *P. paltum* allow perfect correlation with the Paltum and Semicelatum subzones of the NW European standard scale (Elmi et al., 1997; Page, 2003). The Serpentinum Zone has been characterized by the succession of the Harpoceratinae genus (*Eleganticeras–Cleviceras* and *Harpoceras*). The lower boundary has been marked by FAD of *Eleganticeras elegantulum* (level 78 of Ariño). This species is not present at Sierra Palomera, where the FAD of *Harpoceras serpentinum* (level 200) has been used. The Serpentinum Zone has

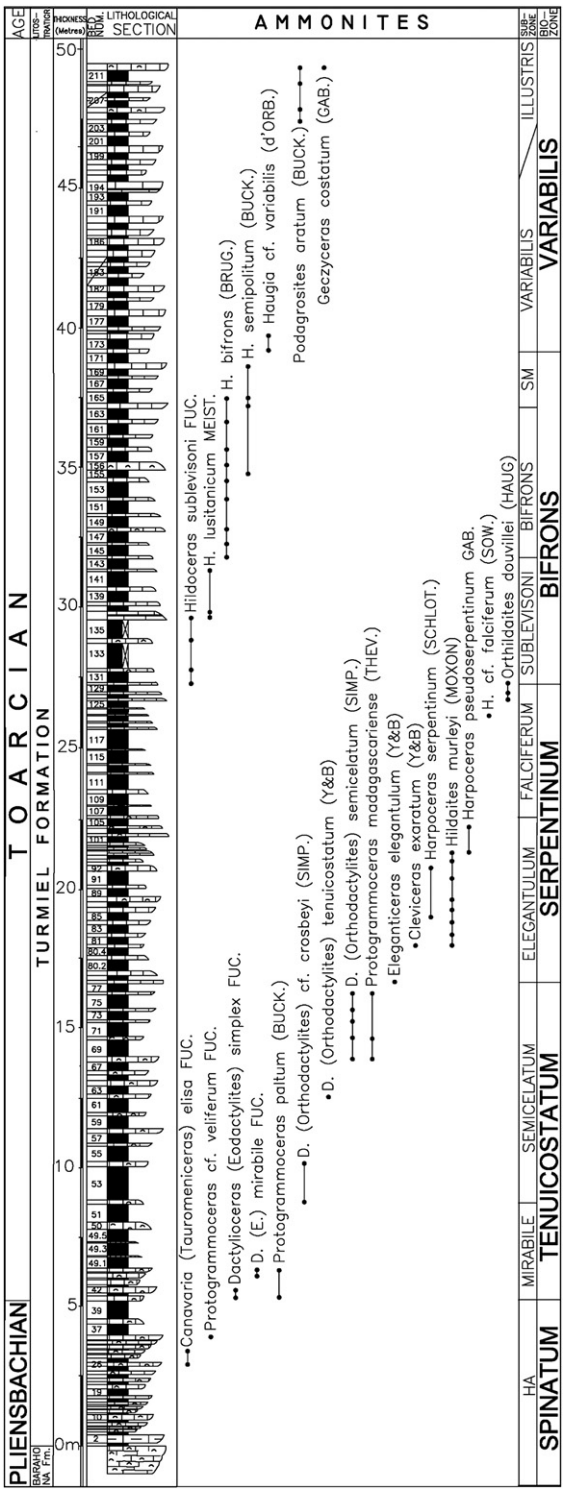


Fig. 4. Stratigraphic succession of the Ariño section showing the distribution of ammonites index characterizing the zones and subzones. These zones and subzones have been referred to the standard chronozones accepted in Europe. Abbreviations: same convention as in Fig. 3. Inclined line indicates the uncertainty interval of the zones and subzones boundaries.

been subdivided into the Elegantulum and Falciferum subzones using the same criteria as the one used in the standard zonation (Elmi et al., 1997).

The lower boundary of the Middle Toarcian has been marked with the FAD of the *Hildoceras* genus. During the Bifrons Zone, the succession of the *Hildoceras* species: *sublevisoni–lusitanicum–bifrons–semipolitum*, allow characterization of the Sublevisoni and Bifrons subzones.

The Variabilis Zone is marked by the FAD of the genus *Haugia*. Within this one, the species of *Haugia: variabilis–illustris–phillipsi–vitiosa*, allow characterization of the Variabilis, Illustris and Vitiosa subzones of the standard scale. In this sector of the Iberian Range, *Haugia illustris* is uncommon. Consequently the lower boundary of the Illustris Subzone has been marked with the FAD of *Podagrosites aratum* species whose FAD in the all basin is slightly younger than the FAD of *H. illustris*. This is based on correlation with other Iberian sections. For instance, in the in the La Almunia-Ricla section the FAD of *H. illustris* is recorded three limestone–marl couplets, or 1.4 m, below the FAD of *P. aratum*, which is present in the Ariño section (Goy and Martínez, 1990).

4. Palaeomagnetic sampling and methods

The magnetostratigraphic sampling has been carried out using a gasoline-powered drilling machine. A magnetic compass and an inclinometer have been used for sample orientation. Deposits corresponding to the Early Toarcian Tenuicostatum and Serpentinum zones, as well as the deposits corresponding to the Middle Toarcian Bifrons and Variabilis zones have been sampled. At Sierra Palomera, 81 oriented cores have been collected along 28 m of section. A preliminary magnetostratigraphic study of this section was carried out by Galbrun et al. (1988b), but the study had not enough resolution to obtain a detailed magnetostratigraphic succession.

In the Ariño section, 74 oriented cores have been collected in a 45 m thick section. The natural remanent magnetisation (NRM) of the samples was measured using a three-axis 2G-Enterprise cryogenic magnetometer. Stepwise acquisition of isothermal remanent magnetisation (IRM) followed by isothermal remagnetisation in three orthogonal directions and subsequent progressive thermal demagnetisation has been carried out in order to identify the magnetic mineralogy of the samples (Lowrie, 1990). The field applied along the three orthogonal axes was: 1000 mT, 400 mT and 100 mT.

Thermal demagnetisation was used as a routine magnetic cleaning technique because it is the only effective method in removing secondary NRM components found

		Standard Subboreal Province (Page, 2003)		Standard NW Europe (Elmi et al., 1994, 1997)		Iberian Range (This work)		Betic Cordillera (Braga et al., 1982; Goy et al., 1988)		Alpine-Apenine (Gallitelli, 1969; Wiedenmayer, 1980)		Standard Mediterranean (Elmi et al., 1994, 1997)	
		ZONES	SUBZONES	ZONES	SUBZONES	ZONES	SUBZONES	ZONES	SUBZONES	ZONES	SUBZONES	ZONES	SUBZONES
TOARCIAN	MIDDLE	VARIABILIS	Vitiosa	VAR.	Vitiosa	VAR.	Vitiosa	GRADATA	Alticarinatus	ERBAENSE	Sternale	GRADATA	Alticarinatus
			Illustris		Illustris		Illustris		Gradata		Gemma		Gemma
			Variabilis		Variabilis		Variabilis		Gemma				
		BIFRONS	Crassum	BIF.	Bifrons	BIF.	Semipolitum	BIFRONS	Semipolitum	BIFRONS	Semipolitum	BIFRONS	Bifrons
			Fibulatum				Bifrons						
			Comune		Sublevisoni		Sublevisoni		Sublevisoni		Sublevisoni		Sublevisoni
	EARLY	SERPENTINUM	Falciferum	SER.	Falciferum	SER.	Falciferum	SER.	Levisoni	FALCIFER		LEVISONI	Falciferum
			Exaratum		Elegantulum		Elegantulum		Striatus				Levisoni
		TENUICOSTATUM	Semicelatum	TEN.	Semicelatum	TEN.	Semicelatum	POLYMORPHUM	Semicelatum	TEN.		POLYMORPHUM	Semicelatum
			Tenuicostatum										
			Clevelandicum										
			Paltum		Paltum		Mirabile		Mirabile				Mirabile

Fig. 5. Correlation chart showing the standard and local ammonite zones and subzones of different areas of Europe (Gallitelli Wendt, 1969; Wiedenmayer, 1980; Braga et al, 1982; Goy et al., 1988; Goy and Martínez, 1990; Elmi et al., 1994,1997; Page, 2003).

in the Iberian Jurassic limestones (Juárez et al., 1998). After a pilot study, all samples were heated from room temperature up to 600 °C or 675 °C in intervals varying between 25, 50 and 100 °C.

5. Magnetic mineralogy and NRM analysis

The IRM acquisition curves indicate major contributions from low coercivity minerals but substantial amounts of high coercivity minerals can be also observed (Fig. 6a). The inflexion of the thermal demagnetisation curve at 575 °C, observed in the low coercivity component (Fig. 6b), indicates the presence of magnetite. Maximum unblocking temperatures of 675 °C of the high coercivity component suggest the presence of haematite in most samples. Contribution of haematite component to IRM experiments varies from sample to sample but is always less than magnetite.

The initial NRM values range between 1.5×10^{-3} and 1.5×10^{-4} A/m. Samples from both sections behave similarly during thermal demagnetisation (Fig. 7 a–d). A present day field component was eliminated during the first demagnetisation steps (commonly up to 150–250 °C). An intermediate component (A component) with maximum unblocking temperatures of 450–475 °C and a high temperature component (B component) were isolated up to 575 °C. Both components are probably carried by magnetite, but the different grain size distributions give distinct spectra of unblocking temperatures (Juárez et al., 1998). This allows good estimates of the direction of each magnetisation component.

In many samples, the initial susceptibility increased after heating to 550–575 °C due to the formation of new mineral phases. This susceptibility increase was often associated with scattering of the magnetisation directions. Therefore, haematite component unblocking at very high temperatures could not be determined.

The mean directions of the A and B components calculated for each section are reported in Table 1 and plotted in Fig. 8. Fold test and reversal test results are summarized in Table 2 and shown in Figs. 8 and 9.

The A component is characterised by always normal polarity and is, on the basis of previous palaeomagnetic research (Juárez et al., 1994, 1998), considered as a remagnetisation component. Similar secondary components have been observed in most Jurassic limestones from the Iberian Range (Juárez et al., 1998). The grouping of the

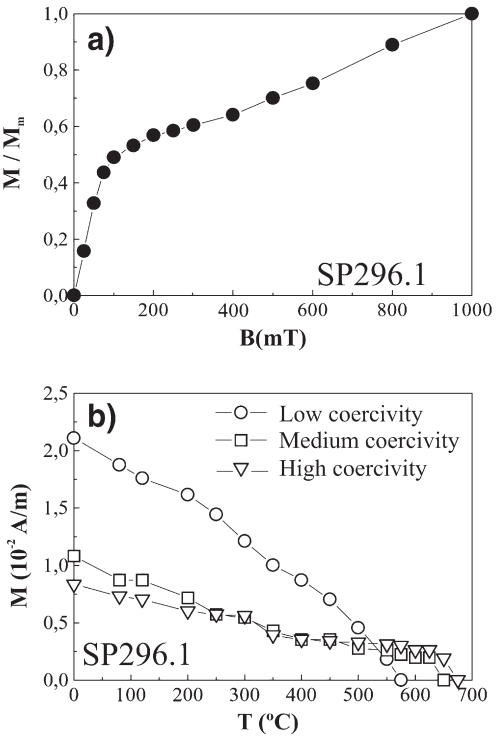


Fig. 6. IRM experiments of a representative sample. (a) IRM acquisition curve and (b) thermal demagnetisation curve of three orthogonal IRM components.

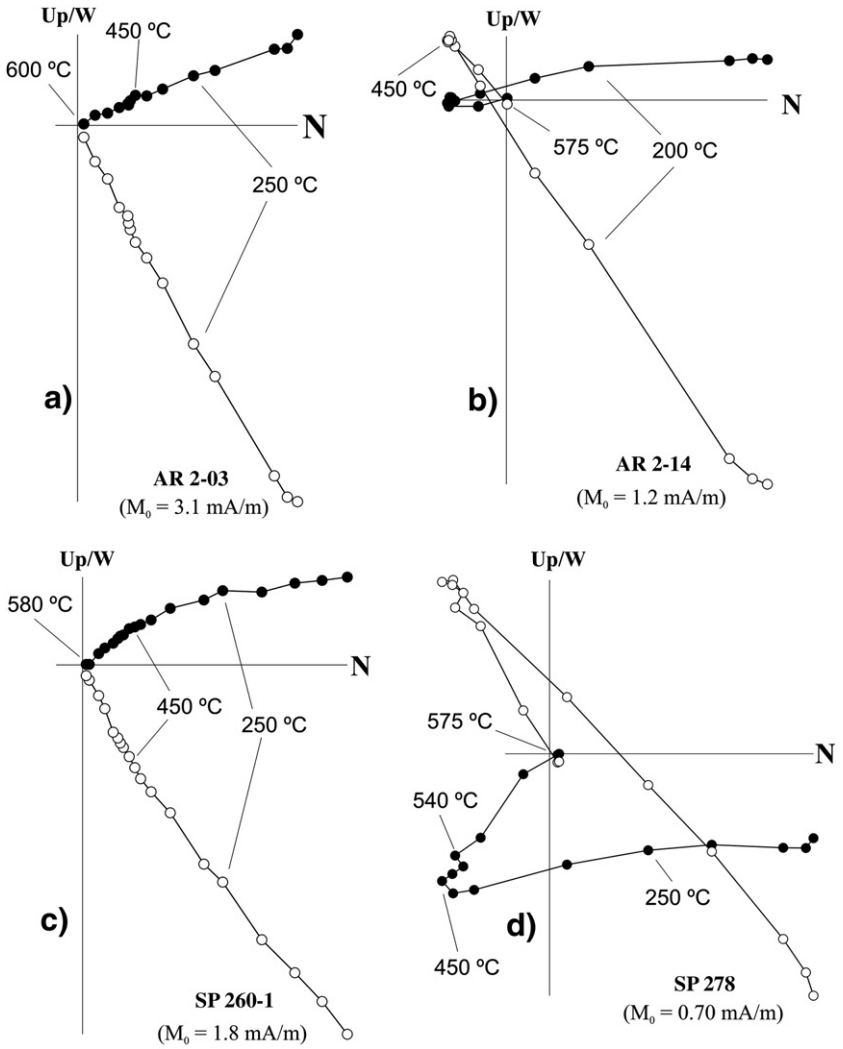


Fig. 7. Orthogonal vector diagrams during thermal demagnetisation of two typical samples from each site (a–b) Ariño (AR), (c–d) Sierra Palomera (SP). (a, c) normal polarity, (b, d) reversed polarity. Closed and open symbols: projections before bedding correction onto horizontal and vertical planes, respectively.

A directions does not improve after nor before tilt correction. This is taken as evidence of a syntectonic origin of this secondary component. An incremental fold test has been performed showing that the best grouping for the A component was achieved after applying 35% of bedding correction (Table 2). The direction of the A component is consistent with the remagnetised component observed in Oxfordian limestones by Juárez et al. (1998). The pole for the remagnetised component is at: Plat=72.2°, Plon=250.8°E (dm=3.3°, dp=3.7°).

The B component, characterized by both positive and negative polarities, has been considered to represent the characteristic remanent magnetisation (ChRM) component of primary origin. This component could not be determined in the lower part of the sections (Mirabile Subzone of the Tenuicostatum Zone) and in the upper part of the Sierra Palomera section (upper part of the Variabilis Zone) due to strong chemical changes produced when heating these samples up to 400–500 °C. The fold test performed on the B component is positive at the 95% level of confidence. The reversal test is positive in both sections: in Sierra Palomera at the 95% and in Ariño at 99% level of confidence. This means that the B component was well isolated.

The palaeomagnetic pole obtained from the ChRM is at Plat=77.4°, Plon=241.3°E (dm=5.4°, dp=6.0°). It represents the first reliable Toarcian palaeomagnetic pole for Iberia.

6. Magnetostratigraphy

The magnetostratigraphic zonations at Sierra Palomera and Ariño are well defined and a good correlation between

Table 1
Palaeomagnetic directions and statistical parameters

Site	n	Dec before	Inc before	K before	α_{95} before	R_{sum} before	Dec after	Inc after	K after	α_{95} after	R_{sum} after
SP											
SP(A)	69	322.7	50.5	48.0	2.5	67.583	357.2	39.3	36.8	2.9	67.152
SP (B) N	43	303.9	54.4	11.3	6.8	39.295	350.4	52.2	12.0	6.6	39.497
SP (B) R	22	114.9	−50.5	20.2	7.1	20.959	163.9	−50.0	16.4	7.9	20.721
SP (B) N+R	65	300.6	53.1	13.2	5.0	60.161	348.1	51.5	13.3	5.0	60.179
AR											
AR(A)	54	351.9	54.3	25.1	3.9	51.891	330.5	61.6	24.9	4.0	51.874
AR (B) N	38	357.5	55.1	14.8	6.2	35.503	348.5	60.2	13.1	6.7	35.168
AR (B) R	24	167.9	−42.3	8.4	10.9	21.256	159.8	−48.4	8.1	11.1	21.146
AR (B) N+R	62	353.3	50.4	10.8	5.8	56.350	344.6	55.8	10.1	6.0	55.988
AR+SP											
SP+AR (A)	123	334.7	53.0	24.5	2.6	118.023	348.7	49.6	16.6	3.2	115.642
SP+AR (B)	127	326.9	54.8	8.4	4.6	111.998	346.5	53.6	11.5	3.9	116.058

SP: Sierra Palomera section; AR: Ariño section; A: A component; B: B component, N: normal polarity; R: reversed polarity; n: Number of specimens; Dec. Inc. Declination and Inclination (before: in geographic coordinates; after: bedding tilt corrected); K and α_{95}: Statistical parameters (Fisher, 1953); R_{sum}: resultant.

both sections can be established (Fig. 10). Internal consistency has been observed between magnetic data, biostratigraphic data and transgressive–regressive cycles. Three normal and three reversed polarity magnetozones have been identified. Within these magnetozones, few magneto-subzones identified in only one section are indicated by lower case letters. All the magnetozones and magneto-subzones are defined by more than one sample, with two exceptions which are marked by a discontinuous bar.

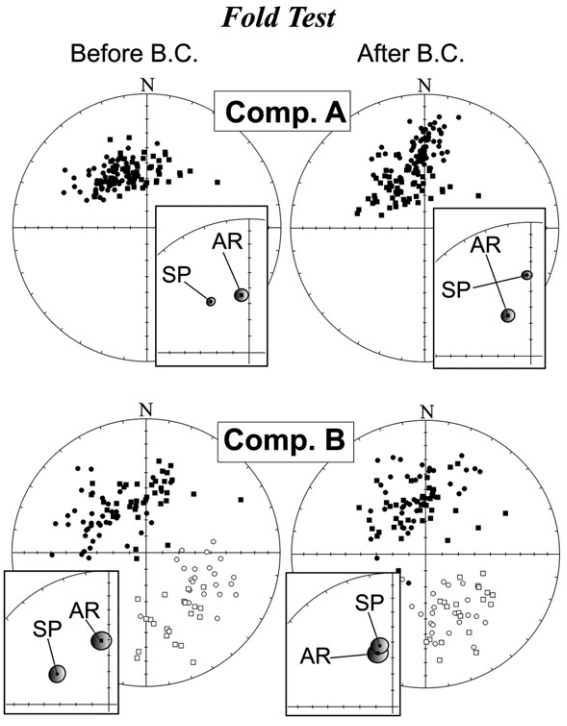


Fig. 8. Stereographic projections of the A and B component directions at Sierra Palomera and Ariño sections before and after tilt correction. Closed and open symbols: positive (lower hemisphere) and negative (upper hemisphere) inclination, respectively. Insets show site mean directions.

The N1 magnetozone covers the upper part of the Tenuicostatum Zone and is well recorded in both sections. The end of this magnetozone is located within the Serpentinum Zone, Elegantulum Subzone. The lower limit of the N1 could not be established because the ChRM could not be determined in the lower part of both sections. R2 is a reversed interval characteristic of the Serpentinum Zone. We named this magnetozone R2 instead of R1 to facilitate the comparison with other magnetostratigraphic studies discussed in the next section.

The N2 magnetozone starts in the lower part of the Falciferum Subzone of the Serpentinum Zone and ends in the middle part of the Bifrons Subzone of the Bifrons Zone. Within N2 interval a reversed magneto-subzone rN2 has been observed in the Sierra Palomera section. Due to a sampling gap in the Ariño section, rN2 and the boundary between the R2 magnetozone and N2 could not be observed.

The R3 magnetozone expands from the Bifrons Subzone up to the Variabilis Subzone. Within R3, a normal magneto-subzone (nR3) is defined in the Sierra Palomera section which seems to be recorded by one sample in the Ariño section. The N3 normal polarity magnetozone located within the Variabilis Zone is well observed in the Ariño section. In the Sierra Palomera section only the lower part of this magnetozone could be determined: the ChRM could not be isolated due to the

Table 2
Fold test and reversal test results

Fold test

Site/Component	n	Before bedding correction					After bedding correction						Fold test result
		Dec.	Inc.	K	α_{95}	f	Dec.	Inc.	K	α_{95}	f	$F_{95\%}$	
Component B SP+AR	127	326.9	54.8	8.4	4.6	0.4219	346.5	53.6	11.5	3.9	0.0101	0,0246	Pre-folding 95% level of confidence
Component A SP+AR	123	334.7	53.0	24.5	2.6	0.4090	348.7	49.6	16.6	3.2	0.8394	0.0251	Syn-folding
↳	COMP A after 35% unfolding:						339.6	52.8	30.2	2.4	0.1125	0.0251	Best clustering distribution after 35 % unfolding, but no statistical confidence is obtained

Reversal test

	Polarity	n	Dec.	Inc.	K	α_{95}	γ_0	γ_c (95%)	γ_c (99%)	Reversal test result
SP (Component B)	N	43	350.4	52.2	12.0	6.6	4.6	10.6	13.3	Positive-C 95% level confidence
	R	22	343.9	50.0	16.4	7.9				
AR (Component B)	N	38	348.5	60.2	13.1	6.7	12.8	12.1	15.1	Positive-C 99% level of confidence
	R	24	339.8	48.4	8.1	11.1				

Same symbols as in Table 1; *f*: fold test statistical parameter (McFadden and Jones, 1981); $F_{95\%}$ significance level value of *f*; γ_0, γ_c (95%) and γ_c (99%), are statistical parameters of reversal test (McFadden and McElhinny, 1990).

already mentioned chemical changes produced by the heating. In the Ariño section, a final reversed magnetozone, starting in the transition between the Variabilis and Illustris subzones, is observed (R4).

7. Discussion

The resulting Ariño-Sierra Palomera composite section spanning the Early–Middle Toarcian has been correlated with other magnetostratigraphic sequences defined in four coeval European/Tethyan pelagic sections: Breggia (Horner and Heller, 1983), Thouars and Airvault (Galbrun et al., 1988a), Vardorbia (Channell et al., 1984), and Iznalloz (Galbrun et al., 1990). In addition, the Iberian composite section has been compared with the Neuquén section from Argentina (Iglesia-Llanos and Riccardi, 2000). This correlation is shown in Fig. 11.

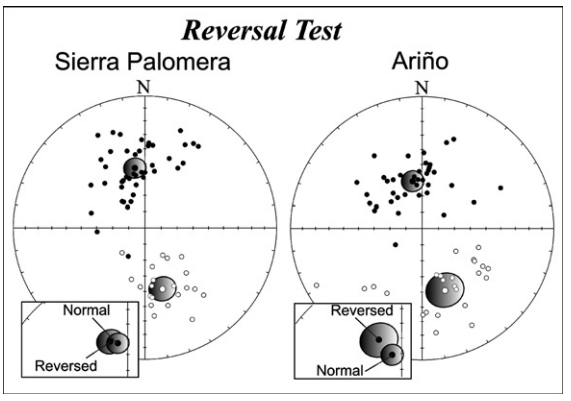


Fig. 9. Stereographic projections of the ChRM (B component) at Sierra Palomera and Ariño showing the reversal test results. Closed and open symbols: positive (lower hemisphere) and negative (upper hemisphere) inclination, respectively. Insets show normal and reversed mean directions for each site.

When comparing the magnetostratigraphic columns one should note that the Ariño and the Sierra Palomera sections are much more expanded than other studied European sections. At Thouars and Airvault, the time interval studied here (Lower–Middle Toarcian) is represented by 1.4 and 1.8 m thick sections respectively, while it is represented by 8.2 m in the Breggia gorge, by 11 m in the Vardorbia section and by 12.5 m in the Ammonitico Rosso facies of Iznalloz. In contrast, the time-equivalent sections studied in the Iberian Range are 37–24 m thick. The Neuquén sections are very expanded (thickness: 225 m), but the sampling was not performed at high resolution (mean sampling distance was about 10 m). In addition, the biozonation in the Argentinean section is partly based on endemic East Pacific faunas, which still implies some difficulties when correlating with the European Standard Zonation. The presence of turbidites brings an additional complexity about the study of the fossils contained in these deposits. Therefore the Sierra Palomera and the Ariño magnetostratigraphic sections should provide the best record of reversals of the earth magnetic field for the Early–Middle Toarcian.

Taking into account the available European sections, a more refined MPTS is proposed in this work (Fig. 11). The magnetozones included in the new MTPS are observed at least in two sections. To be on the safe side, details only

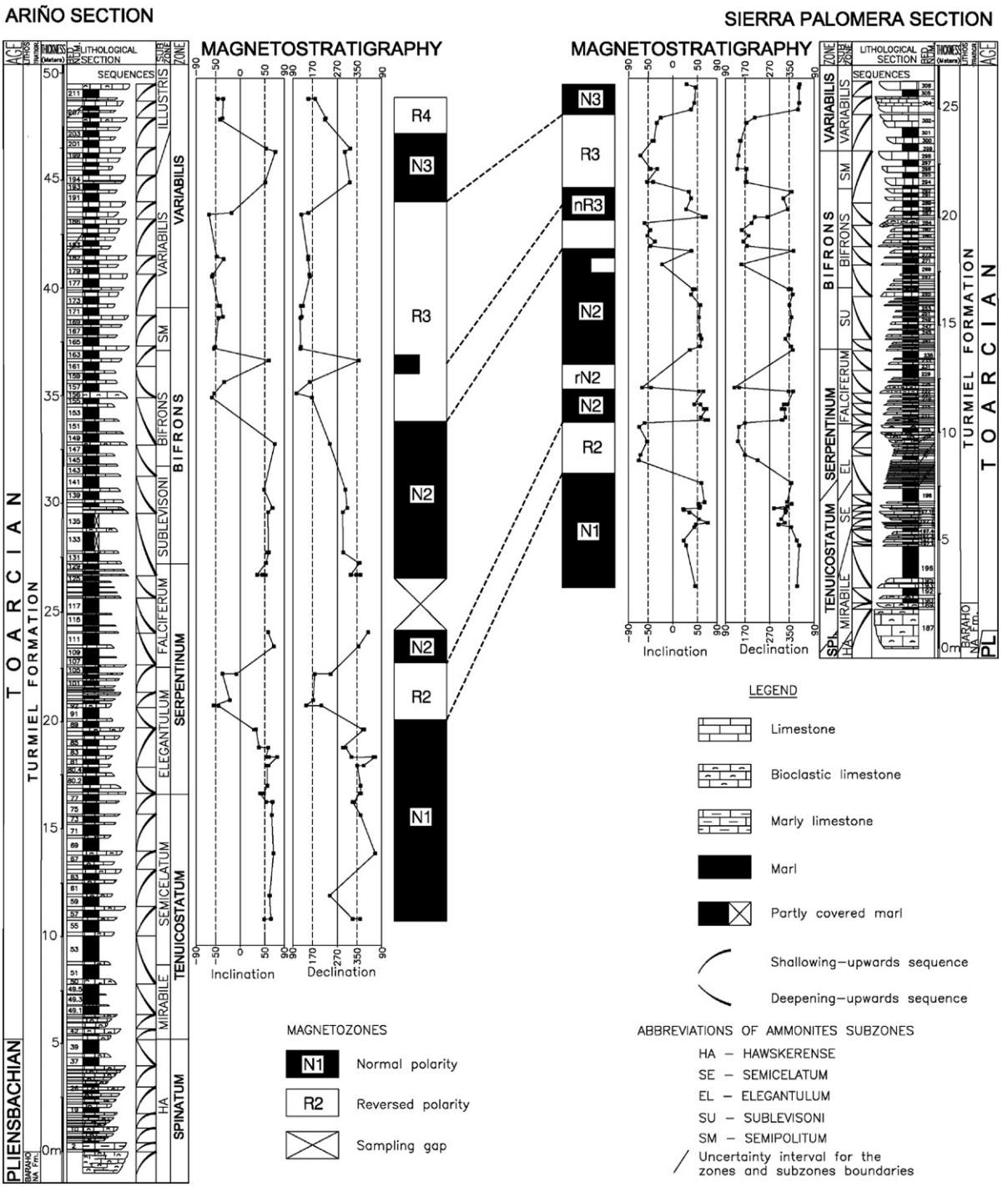


Fig. 10. Declination and inclination values and magnetostratigraphic columns for Sierra Palomera and Ariño sections. Abbreviations: same convention as in Fig. 3.

observed in one section have not been included. In addition, we denominated "magnetozones" when they were present in most sections, and "magneto-subzones" when they were observed in some but not in all sections (indicated by lower case letters). The standard stratigraphic zones and subzones of the Western Europe have been arbitrarily assigned equal duration in order to build up the MPTS.

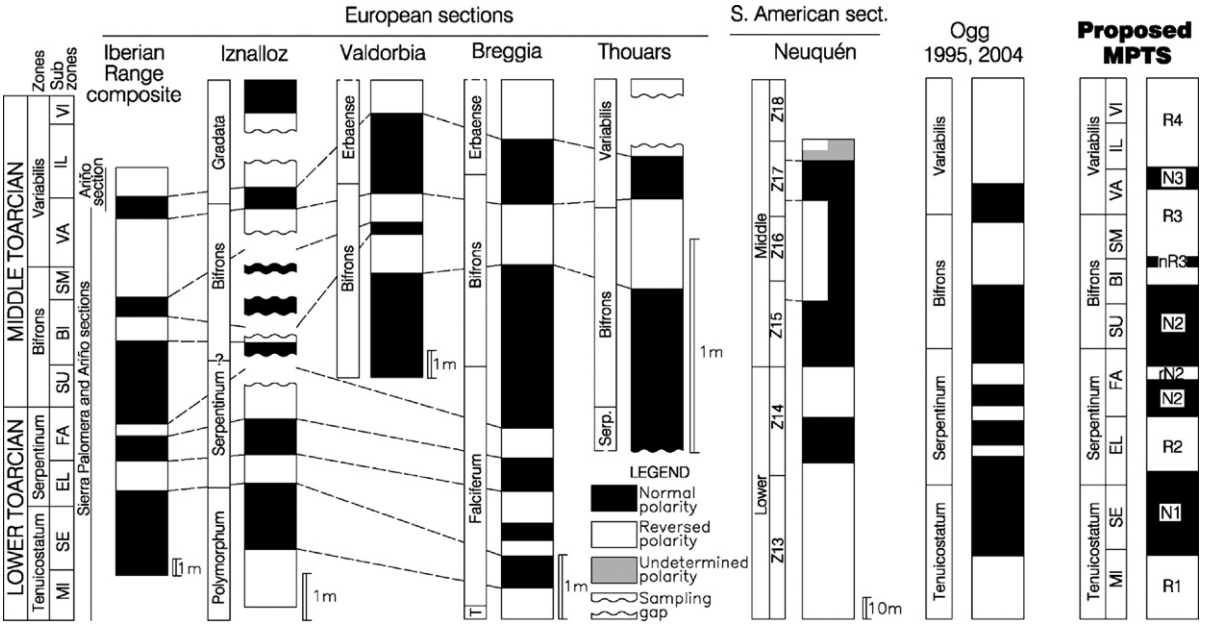


Fig. 11. Left: Magnetostratigraphic correlation between all Toarcian sections and the new MPTS proposed in this paper. MI: Mirabile. SE: Semicelatum. EL: Elegantulum. FA: Falciferum. SU: Sublevisoni. BI: Bifrons. SM: Semipolitum. VA: Variabilis. IL: Illustris. VI: Vitiosa. Right: Comparison between the Toarcian MPTS proposed by Ogg (1995a,b, 2004) and the MPTS proposed in this work.

Correlation between the proposed composite Ariño-Sierra Palomera magnetostratigraphic section and the European sections is quite good. There are, however, some differences, mainly in the Tenuicostatum and Bifrons Zones.

The Breggia section is the most complete magnetostratigraphic section for the Early Jurassic, but the Tenuicostatum Zone (Early Toarcian) is only poorly represented (around 30 cm thick). In the magnetostratigraphic sections at Thouars and Valdorbia the Tenuicostatum Zone is absent. Consequently the best records of the geomagnetic field reversals for the Early Toarcian are the Betic Iznalloz section and the Iberian Range sections studied here.

Most of the Tenuicostatum Zone (partially equivalent to the Polymorphum Zone in the Betic section) is of normal polarity (N1, Fig. 11). The reversed polarity zone observed at the base of the Early Toarcian in Iznalloz and Breggia could not be determined in the Sierra Palomera and the Ariño sections due to unsuitable magnetic properties of samples. Because this reversed magnetozone was observed in the two sections mentioned, we keep this reversed interval in our proposed MPTS. The boundary between the R1 and N1 magnetozones should be located within the Mirabile Subzone according to the Sierra Palomera data.

R2 corresponds to a reversed magnetozone located in the Serpentinum Zone (Falciferum Zone in Breggia). Within R2, one normal polarity zone was only observed in the Breggia section. Magneto-subzones observed in only one section were cautionary not included in our MPTS.

The remaining magnetozones have been observed in all European sections. The exception is the nR3 magneto-subzone that has been observed in our sections and correlated with the Valdorbia and the Iznalloz sections. In spite of the gaps of the Iznalloz section a clear normal polarity chron is detected within the Bifrons Zone.

The apparent offset for the beginning of N3 between the Iberian range composite section and the Iznalloz section could be due to the fact that the onset of the Gradata Zone in the Betic Ranges is assumed to be younger than the Variabilis Zone in NW Europe (Gradstein et al., 1995; Elmi et al., 1989). The age of N3 coincides with the equivalent magnetozone recorded in Thouars (Variabilis Subzone) (Galbrun et al., 1988a). However, the reverse interval R3 in Thouars is considerable thinner (about 30 cm) than in the Ariño section (about 10 m thick) or the Sierra Palomera section (about 3 m thick). The Variabilis Subzone in the Thouars section is condensed and constituted by ferruginous oolitic facies which can contain stratigraphic discontinuities. In addition there is a sampling gap in the Illustris Subzone of the French sections.

N3 is recorded in all sections, but it seems to be diachronic: younger in the Mediterranean sections than in the Western European sections. As reversals are a global

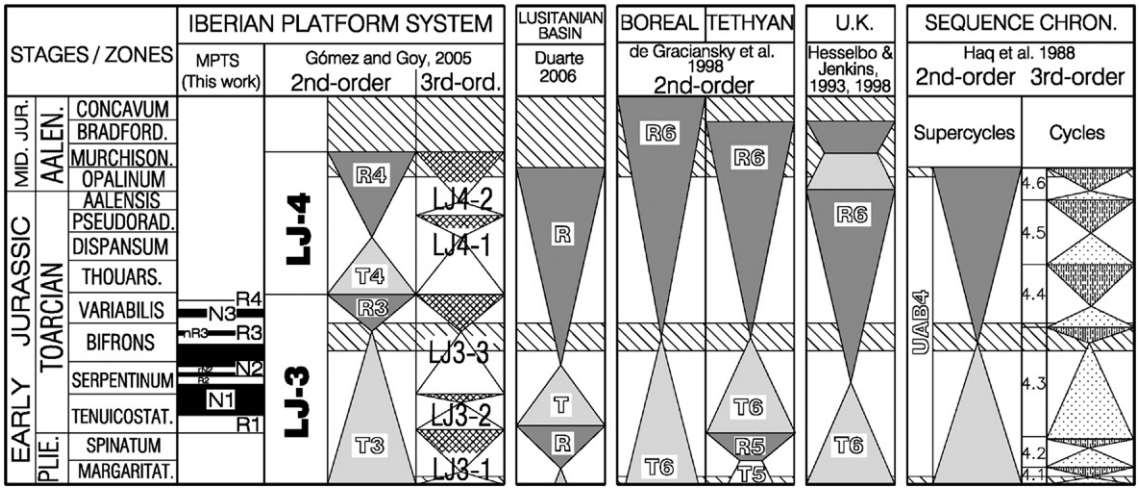


Fig. 12. Correlation chart of the proposed MPTS and the transgressive–regressive facies cycles recognized in the Iberian platform system and observed in other platforms of Europe as well as the 2nd and 3rd order sequence chronostratigraphy (Haq et al., 1988; Hesselbo and Jenkins, 1993, 1998; de Graciansky et al., 1998; Gómez and Goy, 2005; Duarte, 2006).

and synchronic phenomenon, it is more likely that this inconsistency reflects a correlation problem between the NW European and Mediterranean standard chronozones.

Correlation between the new MPTS and the Neuquén section (Iglesia-Llanos and Riccardi, 2000) remains difficult. The length of the sampling interval in the Argentine sections was too high to adequately resolve the sequence of reversals of the earth's magnetic field. The long reversed magnetozone within the Tenuicostatum Zone in the composite Neuquén section is based on three samples from the well dated "AL" section, where the Tenuicostatum Zone is 20 m thick. These results were combined with 4 reversed samples obtained from 75 m of the "RP" section, where the Tenuicostatum Zone was not well defined. Different polarities were observed within Z16 subzone between two sections. Therefore the proposed magnetostratigraphy is very weak because it is based on poor and scarce data. In summary, the Argentinean sections have a good potential to obtain a very detailed record of the reversals of the geomagnetic field in the future if biostratigraphical and taphonomical problems are solved and if a more detailed magnetostratigraphic study is performed. However, at present, they cannot be used for worldwide MPTS comparison.

The resulting proposed MPTS spanning the Early–Middle Toarcian has been correlated and compared with the MPTS proposed by Ogg (1995a,b, 2004) (Fig. 11), which are based in the same magnetostratigraphic information. Differences in previous MPTS (Ogg, 1995a,b, 2004) are only the time scale. For comparison we have rescaled the MPTS proposed by Ogg (1995a,b, 2004) according to our criteria: the standard stratigraphic zones and subzones of the Western Europe have been arbitrarily assigned equal duration.

Correlation between both proposals is quite good. There are, however, some differences: i) There is one magneto-subzone of normal polarity within R2 magnetozone in the MPTS postulated by Ogg (1995a,b, 2004), which is based only on data from the Breggia section (Horner and Heller, 1983). In contrast, no magneto-subzones of normal polarity within R2 are proposed in our MPTS. ii) One magneto-subzone is proposed in our MPTS (nR3) based on data from Sierra Palomera, Ariño, Iznalloz and Valdorbia that was absent in previous MPTS. And iii) the base of the N3 magnetozone is within the Variabilis Zone (Variabilis Subzone) according to the new data presented here and the Thouars section (Galbrun et al., 1988a).

The proposed MPTS will be of great relevance for the correlation of important geological events that took place during the Toarcian, such as timing of mass extinctions, anoxic events, warming and perturbations of the carbon cycle events, as well as local or global sea level changes. In this context, the obtained MPTS has been compared with the transgressive–regressive facies cycles recognized in the Iberian platform system, as well as the cycles reported in other European basins (Fig. 12). Resolution provided by the magnetostratigraphy is higher than the resolution of the 2nd order cycles. Combination of both techniques together represents a powerful tool for global correlation with other areas. One example is represented by the transgressive peak recorded in the Bifrons Zone of most European sections. This is one of the most widespread peak transgressions in the Mesozoic, probably due to a pronounced eustatic sea-level rise. This transgressive

peak coincides in the Iberian platform system with the nR3 magneto-subzone, providing an outstanding correlation horizon that could be of global extent.

8. Conclusions

The well-constrained continuity of the Sierra Palomera and Ariño sections, their high sedimentation rate (compared with the other studied European Toarcian sections), their well-suited palaeomagnetic properties and the excellent mutual correlation, makes the Iberian sections the best candidates for magnetostratigraphic type section for the Early–Middle Toarcian.

Correlation of Iberian results with other Toarcian magnetostratigraphic studies from European sections has produced a more refined MPTS for this period which is proposed in this work.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.epsl.2007.04.048.

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